



Bluesmart luggage

New smart luggage with proximity trackers alert you if you walk away from it
<http://dgit.in/BluesmartBag>



Cardboard desk

Refold is building a standing desk on Kickstarter that is made of cardboard and is pretty strong
<http://dgit.in/RefoldDesk>

of propellant. The tank's pressurisation was achieved using high-pressure Helium allowing Mixed Oxides of Nitrogen to act as oxidizers for the Unsymmetrical Dimethyl Hydrazine fuel. This combination of fuels, with the indigenous design, allowed for long burn times and high integrity of the engines. The spacecraft also had a unified, deployable solar array to generate power for the various systems, payload instruments and energy storage for emergency needs. The solar array was designed in three parts, each of which measured 1.4 meters by 1.8 meters and provided 840 Watts of electrical power while at the distance of Mars from the Sun.

The key to the whole launch system was the system of motors used in the propulsion system. For the Mars mission, ISRO once again didn't see the need to design something from scratch when they already had a trusted model ready. Using the Liquid



The Chandrayaan design was refit for the purposes of Mangalyaan to save on costs.

Apogee Motor (LAM), which had powered Indian geostationary satellites in the past, as well as Chandrayaan-1. The task at hand was to adapt and prepare the LAM on the Main Propulsion System to spark out of dormancy once the 298 days journey to Mars, from Earth orbit, would end. The Attitude Control System was a simpler matter - consisting of eight 22 Newton thrusters using the same fuel to adjust the roll, pitch and yaw as needed.

The propulsive Attitude Control System was accompanied by four reaction wheels for navigational assistance. The system consisted of two star trackers, gyros and a nine headed approximate Sun sensor. The attitude data was also gathered by an Inertial Reference Unit and Accelerometer



Earth's own gravitational energy was used to launch MOM into its designated trajectory

Package, all of which was centered around a MAR31750 computer processor that managed the command, processing and execution of onboard commands.

The communications on board the spacecraft were managed by two 230 Watt Travelling Wave Tubes (TWTA) which operate in the microwave signal range. The spacecraft also had a 2.2 meters parabolic X-Band reflector High Gain Antenna used for data downlinks and command uplink communications. In addition, the craft also includes a medium gain and low gain antenna system. Telemetry and science data was recorded on two 16 Gb Solid State Recorders. This whole package came in at an easy ₹153 crores, far below international standards of expense for missions of this nature.

Achieving Escape Velocity

Since the Mangalyaan was powered by the significantly less powerful PSLV, there was no way to chart a direct trajectory to Mars - there just wasn't enough thrust. The power would need to come from elsewhere and what better way than using the Earth's own gravity? Using the Oberth effect in its plans, the ISRO decided that Mangalyaan would first ascend to Earth orbit and slowly accelerate its interplanetary trajectory with carefully and precisely timed perigee adjustments to achieve escape velocity. This would launch Mangalyaan into a Trans-Martian Trajectory and see it safely between the planets with minimal fuel.

The plans called for carefully timed boosts in Earth orbit known as orbit raising maneuvers. With five such planned stages, each subsequent rise in orbiting altitude

would accelerate the spacecraft until it reached escape velocity. The thrusters on board would make further corrections to ensure that the craft stayed precisely on its intended trajectory. The first three maneuvers were conducted between 5 to 7 November and proved to be successful, while the fourth one on 10 November missed its intended trajectory. To compensate, a supplementary orbit raising manoeuvre was implemented, which restored the spacecraft to its original path.

Another burn of the thrusters proved necessary to propel the craft to its escape velocity. The seventh and final burn took place on 15 November, it was followed 15 days later by a 23 minute firing of the LAM engines. This move thrust Mangalyaan away from the clutches of Earth's gravity at the escape velocity of over 11 kilometers per second and sent it on its way into a heliocentric orbit towards Mars. Mangalyaan would continue this journey for the next 780 million miles in the cold dead of space until it reached Mars.

Only after 298 days did the LAM engines start up again with a 3 second test fire 41 hours before orbit insertion was planned. At 4:17 A.M. Indian Standard Time ISRO was dispatched a signal via the medium gain antenna that the LAM engines were active and 3 hours and 21 minutes after that, a message was received stating that Mangalyaan was in orbit around Mars. The experiment had proved successful!

"Hypothesis: Innovation and perspiration can displace monetary advantages in the pursuit of science. Experiment: Mars Orbiter Mission. Result: Success. Hypothesis confirmed. QED."